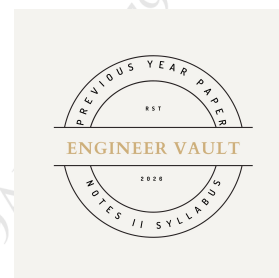


Total No. of Printed Pages: 3

SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group B]Comp., AI & ML,EEE
Examination January/February 2025
Basics of Electrical Engineering

**[Time: 3:00 Hours]****[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION A**Q1** Solve any 10 from the following questions.**20**

1. The total opposition offered to the flow of alternating current by a circuit is called
 - A. Reactance
 - B. Impedance
 - C. Resistance
 - D. Total resistance
2. The ratio of r.m.s. value to the average value of an alternating quantity is known as Ampere
 - A. Form Factor
 - B. Peak Factor
 - C. Maximum Factor
 - D. Current
3. Line voltage in star connected three phase.....times phase voltage.
 - A. 3 time
 - B. $\sqrt{3}$
 - C. Equal
 - D. 2 Times
4. Line current in delta connected three phasetimes line current.
 - A. 3 time
 - B. $\sqrt{3}$
 - C. Equal
 - D. 2 Times
5. The value of form factor of sinusoidal waveform is
 - A. 1.11
 - B. 1.21
 - C. 0.70
 - D. 0.63
6. The unit of electrical field Magneto motive force will be
 - A. AT
 - B. Web
 - C. Ampere
 - D. volt
7. The conductor and flux having angle between them 90 then flux link
 - A. Maximum
 - B. Minimum
 - C. Equal
 - D. Zero

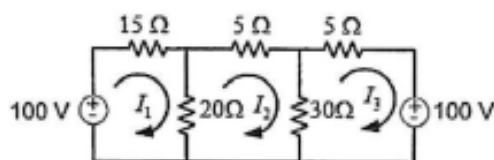
8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known.....
- A. Zero
B. Peak Factor
C. Form Factor
D. None
9. The flux variations being brought about by variations
- A. Density
B. voltage
C. current
D. Power
10. An ideal voltage source is that which has an internal resistance
- A. Zero
B. Low
C. High
D. Infinite
11. The energy used in one hour at the rate of 1kW is known as.....
- A. 1 Unit
B. 1 watt
C. 1 Newton
D. 1 volt
12. In Kirchhoff's first law $\sum I_i = 0$ at the junction is based on the conservation of
- A. Charge
B. Current
C. Energy
D. Voltage
13. The resistance of a material is most commonly determined by four factors- length, cross-sectional area, type of material and
- A. Current
B. Voltage
C. Power
D. Temperature



SECTION B

- Q2** a) Prove the Series and parallel combination of resistor with circuit diagram. **05**
b) Expression for value of α at different temperature in detail. **05**

- Q3** a) Find loop Current I_1 , I_2 & I_3 by Mesh Analysis. **05**



- b) Explain Superposition theorem with an example. **05**
- Q4** a) Comparison between Magnetic and Electric Circuits. **05**
b) A solenoid have 800 turns supply with 30 volt of resistance is 6 ohm, length of solenoid is 16 cm with area is 6 sq.cm, relative permeability is 800 find 1) MMF **05**
2) Magnetic field intensity 3) Flux density 4) Reluctance

- Q5** a) Prove the Coefficient of Coupling with equation. **05**
b) Prove the Capacitors in Series with detail. **05**
- Q6** a) Explain R-L-C Series A.C. Circuit with phasor diagram **05**
b) Explain R-C Series A.C. Circuit with phasor diagram **05**
- Q7** a) Explain & Draw the phasor diagram for Star connected phase voltage and line voltage **05**
b) Explain & Draw the phasor diagram for Delta connected phase current and line current **05**

